

Awesome Articles

I am a regular reader of EFY. The article 'How 3D Glasses Work' published in November issue is really awesome. It gives good information on 3D display technologies and 3D glasses. The article 'Web-based Online Simulator' is useful for beginners, students and hobbyists. The article 'Transformerless Power Supply' explains the design so well that even beginners can assemble it. 'Technology News' and 'New Products' sections are very resourceful. The '1kW Sinewave Inverter' construction project can help readers assemble their own low-cost inverter. Hope to see good articles on microcontrollers' comparison too.

Sunam
Kollam

EFY: Thanks for the feedback and appreciation! We will try to publish an article on microcontrollers' comparison as soon as possible. Learned readers may submit suitable articles on this topic.

LED Lighting

The article 'Future Looks Bright for LED Lighting' by Janani Gopalakrishnan in September issue was very informative. I congratulate the author for the same. However, I feel that a table comparing the efficiencies (in lumen/watts) of currently available LED lamps with incandescent lamps would have been helpful. Also, it is

'Spot An Error' Award Winners

❑ In 'Transformer-less Power Supply: A Cost-effective Power Solution' article in November issue, on page, 48, the reactance value of $0.22\mu\text{F}$ capacitor is mentioned as 14.4 ohms. The actual value should be 14.4 kilo-ohms.

Vighnesh K.S.P.

In the same article, on page 50, 318 mW is wrongly noted as 318 mV in the second last paragraph.

Suraj

❑ In 'Designing with FPGA' article in November issue, on page, 68, VHDL is wrongly described as "very high-scale integrated circuit hardware description language" instead of "very high-speed integrated circuit hardware description language."

Sridhar Bukya

Efficacy and Life of Incandescent vs LED Lighting Components

	40W bulb	7W LED	60W bulb	12.5W LED
Lumens	550	400	900	800
Efficacy (lm/W)	14	57	15	64
Life (hours)	5000	100,000	1000	25,000

stated that LEDs have an operating life of up to 10,000 hours. But what about the life of a complete lamp assembly using SMPS or converters? From the point of the user, the life of complete light assembly is more important.

Dr G.N. Acharya

Pune

The author Janani Gopalakrishnan Vikram replies:

Thanks for the appreciation; it is the feedback from readers like you that

encourages us to keep improving our work.

Answering the query raised you, first, I would like to clarify that the article mentions the life of white LEDs as more than 100,000 hours, not 10,000 hours. Second, the article focuses on LED lighting components. Regarding the efficacy and life of incandescent vs LED lighting components, a rough estimate based on recent reports is shown here in the table (a similar table was published in September 2011 issue of EFY, and not much has changed since then).

These are generally the theoretical estimates of the lifetime of LED lighting components. The practical lifetime of lamps will vary depending on the conditions of use, packaging technology, deterioration in light quality, etc. This information cannot be generalised as it is dependent on application and technology.

Since it is not possible to test LED fixtures for 6-10 years before publicising a figure, there are certain standards used to calculate the average useful life of LED products. If you go with reputed brands, you can check their website for

From Facebook: EFY's Electronics Design Community

❑ In 'Touchscreen Control for Wheelchair' project (by Sachidananda Sahu) in EFY July, the model number of the touchscreen was not mentioned in the article. Can I replace it with a resistive touchscreen, without change in the hardware and software?

If the wheelchair is used, it will require motors with high ratings. Will L293D suffice to handle this?

Nishant Mittal

❑ L293D will not work. My friends have worked on a real chair; they used MOSFET drives. Visit the website of element14 to get suitable motor drivers.

Samuel Noah Ramrajkar

❑ I have used a resistive touchscreen, but you can use any touchscreen by changing the code according to the hardware. About the motor, it's an idea and demo model. For the practical wheelchair, you will have to use high-torque motors and drivers for the wheelchair.

Sachidananda Sahu